Video Number: V27 Video Title: V27 SC4024 Exploring Lifestyle Insights

1. Choice of visuals, visualisation techniques and appropriate use of human visual perception principles

[1] [0:22] Workout Types – Females and Males (Bar chart)

Strengths: The bar charts used here are excellent for categorical comparison (gender vs workout type) showing workout preferences across genders. These two bar charts effectively represent workout types for males and females using categorical colors, red for females and blue for males. Using the Gestalt Principle of Simplicity, the labels are ordered in descending order and in both graphs, the highest and lowest workout types ('Strength' and 'Cardio') are highlighted using the Gestalt Principle of Enclosure.

The pre-attentive color differences red for females and blue for males help in quick comparison while the horizontal layout helps with the readability. The charts align with the narration, establishing a clear visual hierarchy.

Areas for Improvement: Although each chart is individually clear and easily readable, comparing both simultaneously is difficult because of different order of workout type labels. To convey the message effectively and enhance the comparison, either a grouped bar chart or butterfly chart would allow to display both genders side-by-side for each workout type, thus, helping with quick comparison.

Also, using distinct highlighting colors for the enclosed labels (e.g., green for Strength and yellow for Cardio) would have improved pre-attentive differentiation and Gestalt Similarity making it easier to distinguish between highlighted categories improving readability.

Additional Comments: The chart demonstrates strong adherence to HVP principles of color (using distinct colors and labels) and alignment (using common scale) and Gestalt principles (Simplicity and Enclosure) but could better support Similarity (distinct label colors) and Continuity by aligning both gender charts on the same order. Plotting both genders on same grouped or butterfly bar chart would further enhance proximity and continuity allowing for more direct visual comparison.

[2] [0:53] Top 10 Workouts – Popularity by Gender (Grouped Bar Chart)

Strengths: The grouped bar chart effectively shows differences in workout popularity across different exercises and genders. Consistent use of colors with previous chart **[1] [0:22]** maintains narrative continuity and video flow. The chart supports pre-attentive processing of color and length helping with a quick comparison.

Areas for Improvement: The x-axis is overcrowded with rotated x-axis labels which reduces the readability and increases cognitive load. A horizontal orientation of the chart would have improved the readability by aligning with natural reading direction and easier reading of long labels. This would have also enhanced the visual look of chart reducing x-axis crowdedness. The chart also lacks clear ordering violating Gestalt Law of Simplicity. In this case, ordering the workout labels alphabetically would have improved clarity.

Additional Comments: Grouped bar chart is appropriate for bivariate categorical comparison, however, adjusting orientation and labels could further enhance pre-attentive visual perception and Gestalt Law of Simplicity. Accessibility featured like adding texture or pattern could be considered to improve readability for individuals with blue or red color-blindness.

[3] [1:23] Training Frequency (Histogram)

Strengths: The histogram appropriately visualises the distribution of training frequency. The Gestalt Principle of Enclosure highlights the most frequent workouts.

Areas for Improvement: There is crowding on x-axis and it is unevenly labeled. The possible reason could be too many ticks or small chart width. Reducing ticks, widening chart or increasing bin width may help in resolving this issue. The filled area style bar lacks clear boundaries reducing figure-ground separation and making it harder to distinguish individual bars. Using distinct (white) bar boundaries/borders would enhance pre-attentive separation and visual clarity.

Additional Comments: Instead of enclosing most frequent bars (Gestalt Principle of Enclosure), Gestalt Principle of Similarity could have been used here by choosing darker color for higher frequency bars. This would add Principle of Salience (8 Psychological Principles by Stephen Kosslyn) and pre-attentive visual processing.

[4] [1:44] Training frequency distribution by Experience Level (Violin plot with box plot overlay)

Strengths: The combination of violin plot with a boxplot overlay effectively communicates median, outliers and overall distribution. Categorical colors and labels for different experience levels are enhancing readability. The visualisation adheres well to Gestalt Principle of Symmetry contributing to balanced visual layout.

Areas for Improvement: There is lack of annotations (e.g., median). Annotating median would have provided additional information to enhance understanding. Annotating key values directly above or next to categories could improve visual hierarchy and reinforce Gestalt Principle of Continuity by guiding viewer's eye smoothly along the perceived upward trend in training frequency with experience.

Label placement could also be improved. Positioning labels within or beside each violin shape would increase clarity.

Additional Comments: There is an absence of legend (explaining what each color represents). Even if it is intuitive from x-axis labels, redundancy improves accessibility.

[5] [1:59] Correlation between Age, BMI and Workout frequency (Heatmap)

Strengths: Heatmaps are effective for correlation analysis with color intensity mapping to strength of correlation. A diverging palette fits the purpose well with neutral color for no correlation and higher intensity for stronger correlation. Hence, there is pre-attentive color encoding, supporting quick interpretation.

Areas for Improvement: Instead of a blue-red, a green-red diverging palette might be more intuitive where green represents positive and red represents negative correlation aligning better with semantic associations. Alternatively, a blue-orange palette could enhance accessibility for color-blind viewers while maintaining perceptual balance.

Additional Comments: The chart appears too small relative to the screen space, and the color legend is placed too far from the chart, creating split attention and reducing figure-ground clarity. Enlarging chart with closer legend would have balanced it enhancing visual appeal and figure-ground clarity with the Gestalt Principle of Proximity.

[6] [2:11] BMI distribution by Experience Level (Pie Chart)

Strengths: Pie chart is very effective to show composition. Each pie chart individually displays the composition of BMI categories well and the colors are intuitively chosen (e.g. green = acceptable). The categorical colors support semantic color perception and pre-attentive processing.

Areas for Improvement: Each pie chart shows BMI composition well individually however comparing them across different experience levels is difficult. They are perceptually weaker for cross-category comparisons and more cognitive demanding. Additionally, length encoding is more effective and reliant than area and angle (Mackinlay's Visual Encoding Effectiveness Ranking) therefore, a stacked bar chart here would enable more precise comparisons across experience levels using pre-attentive length encoding.

Additional Comments: Using a stacked bar chart with consistent percentage annotations and consistent colors and semantic mapping would strengthen pre-attentive processing and Gestalt Principle of Similarity, reducing cognitive load and conveying the intended message efficiently.

[7] [8] [9] [2:28] – [2.49] Popular Workouts- Beginners, Intermediate, Advanced (Treemap)

Strengths: The treemap efficiently summarises multi dimensional data (workout type and popularity) within a compact space.

A sequential color palette conveys intensity and color saturation conveys popularity effectively using pre-attentive color encoding.

Areas for Improvement: Overuse of vibrant color violates color linearity and causes perceptual imbalance.

Usually darker colors imply higher values. In this chart, blue is representing less popular workouts but blue being darker than yellow is drawing more undue attention than yellow, despite yellow representing most popular workouts.

Instead of using multiple colors, a single hue sequential color palette (for e.g., blue - light to dark) would have looked better with darker color representing higher values. Moreover, many labels are not visible hence consistent font style and size would improve the readability.

Additional Comments: While the charts show rich information, these look visually overloaded taking a lot of space. These appear overwhelming, violating the Principle of Relevance (Goldilocks Principle) and Principle of discriminability (failing to maintain good contrast in colors). A bubble chart could convey the similar insights with clearer encoding (size for frequency, color for intensity) with better usage of Gestalt Principle of Simplicity and two above principles.

[10] [3:22] Workout Duration Across Experience Levels (Line Chart)

Strengths: Line chart effectively conveys upward trend in duration with increasing experience.

Areas for Improvement: For improving the line chart, labeling of beginner, intermediate and advanced on the chart in the direction of line would have helped in conveying the intended message.

Animating the line from left to right would make the message visually clear that there is an upward trend aiding pre-attentive direction perception through the slope direction, thus, improving engagement and Gestalt Principle of Continuity.

Additional Comments: Line chart looks visually sparse. While the line chart suggests a continuous progression of workout duration with experience levels. I assume experience level variable is more likely categorical (beginner, intermediate, advanced). Using a line chart can therefore mislead by implying continuity between discrete groups. A bar chart or box plot would better reflect the categorical nature of the data and make comparisons between experience levels clearer.

[11] [3:42] Calorie Density by Workout Type (Box plot)

Strengths: Box plot effectively illustrates the spread and median calorie burn across different workout types. Using categorical colors support pre-attentive grouping and comparison.

Areas for Improvement: Inconsistent y-axis labeling (missing baseline value) affects interpretability. Reordering the categories with median values would enhance Gestalt law of Simplicity.

Additional Comments: Presence of legends could enhance readability. Even the x-labels are there for each workout type, redundancy enhances accessibility.

[12] [13] [4:03] – [4:26] BMI vs Protein Intake & Calorie vs Fat Percentage by Diet Type (Scatter Plot with Regression Line)

Strengths: The choice of scatter plot is ideal with appropriate colors chosen for different categories and legend at the right with labeling of correlation values. Regression line helps with direction perception clearly by applying pre-attentive slope processing and communicating the trends effectively.

Areas for Improvement: To strengthen visual semantics, the positive correlation chart [13] [4:26] could use green to intuitively represent positive outcomes.

2. Overall Comments

Strongest feature: The strongest feature of the video presentation is the simplicity and clarity of visualisation choices which makes the insights easy to understand. The pacing and the narration are effective, with each visualisation displayed long enough for viewers to absorb key information. There is a logical flow of storytelling from demographics to workout efficiency followed by diet outcomes.

The video uses simple yet effective visualisations with a clear message which helps to understand the message conveyed quickly. A variety of visualisation type such as bar charts, box plots, scatterplots and heatmaps have been used effectively.

Color consistency between gender categories across charts [1], [2] supports Gestalt Continuity and pre-attentive color processing. The Gestalt Principle of Enclosure is applied consistently ([1], [3], [11]) to highlight important insights. The use of appropriate color palette such as categorical [1], [2], [4], [6], [11], sequential [7-9] and diverging [5] is used with semantic mappings ([1],[2], [6]) demonstrating thoughtful visual choices.

Weakest feature: The weakest aspect of the video is the limited use of annotations, interactivity and animation, which reduces the engagement and weakens the narrative flow. For e.g., in chart [10], animating the line chart from left to right could have visually reinforced the upward trend, while clear labelling of experience levels would have enhanced message clarity.

Additionally, some visualisation choices are suboptimal ([6], [1], [7-9] and [10]). There are some inappropriate chart types, limited adherence to perceptual accuracy like color ([5], [7]), scale [3], and clutter ([2], [3]). There is also a weak application of HVP principles and Gestalt principles ([2], [7-9]).

Additionally, a more suitable color palette could have been used in [7-9] to convey the intended message more effectively. Accessibility consideration such as color-blind safe palettes and texture differential were also overlooked (for e.g., [2], [6], [1]).

Overall, while the narrative of video is strong and the visualisations generally clear, there remains room to improve perceptual precision, engagement, and interpretability. Stronger adherence to HVP and Gestalt Principles along with more thoughtful use of annotations, labeling, transitions, and animations would have strengthened the viewer's understanding and overall impact of the presentation.